

DS4002Project2Code

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Load data

```
sea_ice_data<-read.csv("~kfreese/Desktop/DS 4002/Project 2/Code/sea_ice_data_clean.csv")
head(sea_ice_data)
```

```
##   year mo source_dataset extent  area
## 1 1979  1      NSIDC-0051  15.41 12.41
## 2 1980  1      NSIDC-0051  14.86 11.94
## 3 1981  1      NSIDC-0051  14.91 11.91
## 4 1982  1      NSIDC-0051  15.18 12.19
## 5 1983  1      NSIDC-0051  14.94 12.01
## 6 1984  1      NSIDC-0051  14.47 11.68
```

```
View(sea_ice_data)
```

Load packages

```
library(dplyr)      # for data manipulation
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##   filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)    # for plotting
```

```
library(zoo)         # for rolling standard deviation
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
library(forecast)  # for ARIMA modeling and forecasting
library(tseries)   # for the ADF stationarity test
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
library(lubridate) # for working with dates
```

```
##
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union
```

1: Data Cleaning

```
# Read in the cleaned CSV file
sea_ice <- read.csv("sea_ice_data_clean.csv", stringsAsFactors = FALSE)

# View the first few rows
head(sea_ice)
```

```
##   year mo source_dataset extent area
## 1 1979  1      NSIDC-0051  15.41 12.41
## 2 1980  1      NSIDC-0051  14.86 11.94
## 3 1981  1      NSIDC-0051  14.91 11.91
## 4 1982  1      NSIDC-0051  15.18 12.19
## 5 1983  1      NSIDC-0051  14.94 12.01
## 6 1984  1      NSIDC-0051  14.47 11.68
```

```
# View the structure of the dataset
str(sea_ice)
```

```
## 'data.frame':   564 obs. of  5 variables:
##  $ year      : int  1979 1980 1981 1982 1983 1984 1985 1986 1987 1989 ...
##  $ mo        : int   1 1 1 1 1 1 1 1 1 1 ...
##  $ source_dataset: chr   " NSIDC-0051" " NSIDC-0051" " NSIDC-0051" " NSIDC-0051" ...
##  $ extent     : num  15.4 14.9 14.9 15.2 14.9 ...
##  $ area       : num  12.4 11.9 11.9 12.2 12 ...
```

```
# Check dimensions of the dataset
dim(sea_ice)
```

```
## [1] 564 5
```

```
# Check column names
names(sea_ice)
```

```
## [1] "year"          "mo"            "source_dataset" "extent"
## [5] "area"
```

```
# Remove rows with invalid placeholder values if any still remain
sea_ice <- sea_ice %>%
  filter(extent != -9999, area != -9999)

# Check for missing values in each column
colSums(is.na(sea_ice))
```

```
##          year          mo source_dataset      extent      area
##          0           0           0           0           0
```

```
# Check for duplicate rows
sum(duplicated(sea_ice))
```

```
## [1] 0
```

```
# Remove duplicate rows if any exist
sea_ice <- sea_ice %>%
  distinct()

# Keep only rows with nonnegative extent values
sea_ice <- sea_ice %>%
  filter(extent >= 0)

# Rename month column from "mo" to "month" for readability
sea_ice <- sea_ice %>%
  rename(month = mo)

# Create a proper date variable using year, month, and day = 1
sea_ice <- sea_ice %>%
  mutate(date = as.Date(paste(year, month, 1, sep = "-")))

# Sort the data chronologically
sea_ice <- sea_ice %>%
  arrange(date)

# Create a time index variable from 1 to number of rows
sea_ice <- sea_ice %>%
  mutate(t = 1:n())

# Create a decade variable
```

```
sea_ice <- sea_ice %>%
  mutate(decade = floor(year / 10) * 10)

# Verify the cleaned data
head(sea_ice)
```

```
##   year month source_dataset extent  area      date t decade
## 1 1978    11      NSIDC-0051  11.65  9.04 1978-11-01 1  1970
## 2 1978    12      NSIDC-0051  13.67 10.90 1978-12-01 2  1970
## 3 1979     1      NSIDC-0051  15.41 12.41 1979-01-01 3  1970
## 4 1979     2      NSIDC-0051  16.18 13.18 1979-02-01 4  1970
## 5 1979     3      NSIDC-0051  16.34 13.21 1979-03-01 5  1970
## 6 1979     4      NSIDC-0051  15.45 12.53 1979-04-01 6  1970
```

```
tail(sea_ice)
```

```
##   year month source_dataset extent  area      date  t decade
## 559 2025     8      NSIDC-0803   5.39  3.33 2025-08-01 559  2020
## 560 2025     9      NSIDC-0803   4.75  3.08 2025-09-01 560  2020
## 561 2025    10      NSIDC-0803   6.32  4.69 2025-10-01 561  2020
## 562 2025    11      NSIDC-0803   8.77  7.36 2025-11-01 562  2020
## 563 2025    12      NSIDC-0803  11.22  9.60 2025-12-01 563  2020
## 564 2026     1      NSIDC-0803  13.11 11.39 2026-01-01 564  2020
```

Sea ice area = total region covered by ice.

Extent = total region with at least 15% sea ice cover. NSIDC reports extent.

2: Descriptive Analysis

```
# Overall summary statistics for extent
summary(sea_ice$extent)
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3.570   8.477  12.015  11.309  14.200  16.340
```

```
sd(sea_ice$extent)
```

```
## [1] 3.283034
```

```
# Summary statistics by decade
decade_summary <- sea_ice %>%
  group_by(decade) %>%
  summarise(
    mean_extent = mean(extent, na.rm = TRUE),
    median_extent = median(extent, na.rm = TRUE),
    min_extent = min(extent, na.rm = TRUE),
    max_extent = max(extent, na.rm = TRUE),
```

```

    sd_extent = sd(extent, na.rm = TRUE),
    n         = n()
  )

print(decade_summary)

```

```

## # A tibble: 6 x 7
##   decade mean_extent median_extent min_extent max_extent sd_extent      n
##   <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl> <int>
## 1  1970      12.4      12.9      7.05      16.3      3.05    14
## 2  1980      12.2      12.5      6.7       16.1      2.96   117
## 3  1990      11.8      12.4      6.08      15.9      3.04   120
## 4  2000      11.1      11.8      4.27      15.5      3.23   120
## 5  2010      10.5      11.4      3.57      15.2      3.47   120
## 6  2020      10.4      11.2      4         14.9      3.51    73

```

```

# Summary statistics by month
month_summary <- sea_ice %>%
  group_by(month) %>%
  summarise(
    mean_extent = mean(extent, na.rm = TRUE),
    median_extent = median(extent, na.rm = TRUE),
    min_extent = min(extent, na.rm = TRUE),
    max_extent = max(extent, na.rm = TRUE),
    sd_extent = sd(extent, na.rm = TRUE),
    n = n()
  )

print(month_summary)

```

```

## # A tibble: 12 x 7
##   month mean_extent median_extent min_extent max_extent sd_extent      n
##   <int>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl> <int>
## 1     1      14.1      14.2      13.1      15.4      0.626    47
## 2     2      15.0      15.2      13.8      16.2      0.631    47
## 3     3      15.2      15.2      14.1      16.3      0.581    47
## 4     4      14.5      14.5      13.4      15.5      0.563    47
## 5     5      13.1      13.0      11.9      14.1      0.508    47
## 6     6      11.5      11.5      10.4      12.5      0.638    47
## 7     7       9.00      9.21      7.29      10.6      0.966    47
## 8     8       6.66      6.77      4.72       8.19      1.06     46
## 9     9       5.88      6.08      3.57       7.67      1.16     47
## 10    10       7.76      7.93      5.33       9.48      1.19     47
## 11    11      10.4      10.3      8.66      11.6      0.796     48
## 12    12      12.6      12.6      11.2      13.7      0.670     47

```

```

# Create annual average extent
annual_extent <- sea_ice %>%
  group_by(year) %>%
  summarise(
    n_months = n(),
    annual_extent = mean(extent, na.rm = TRUE)
  )

```

```

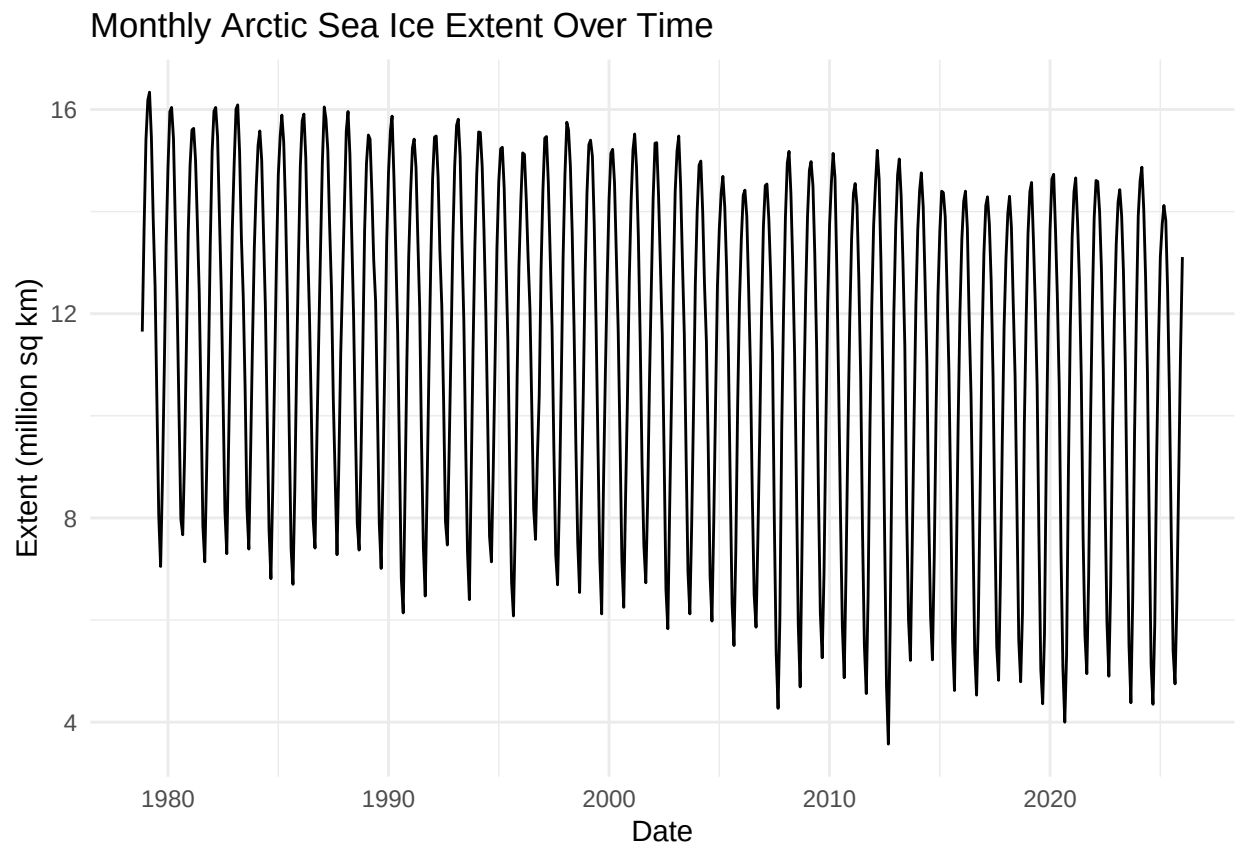
) %>%
  filter(n_months == 12)

# Create September-only dataset
sept_data <- sea_ice %>%
  filter(month == 9) %>%
  arrange(year)

# Create March-only dataset
march_data <- sea_ice %>%
  filter(month == 3) %>%
  arrange(year)

# Plot 1: Monthly sea ice extent over time
ggplot(sea_ice, aes(x = date, y = extent)) +
  geom_line() +
  labs(
    title = "Monthly Arctic Sea Ice Extent Over Time",
    x = "Date",
    y = "Extent (million sq km)"
  ) +
  theme_minimal()

```

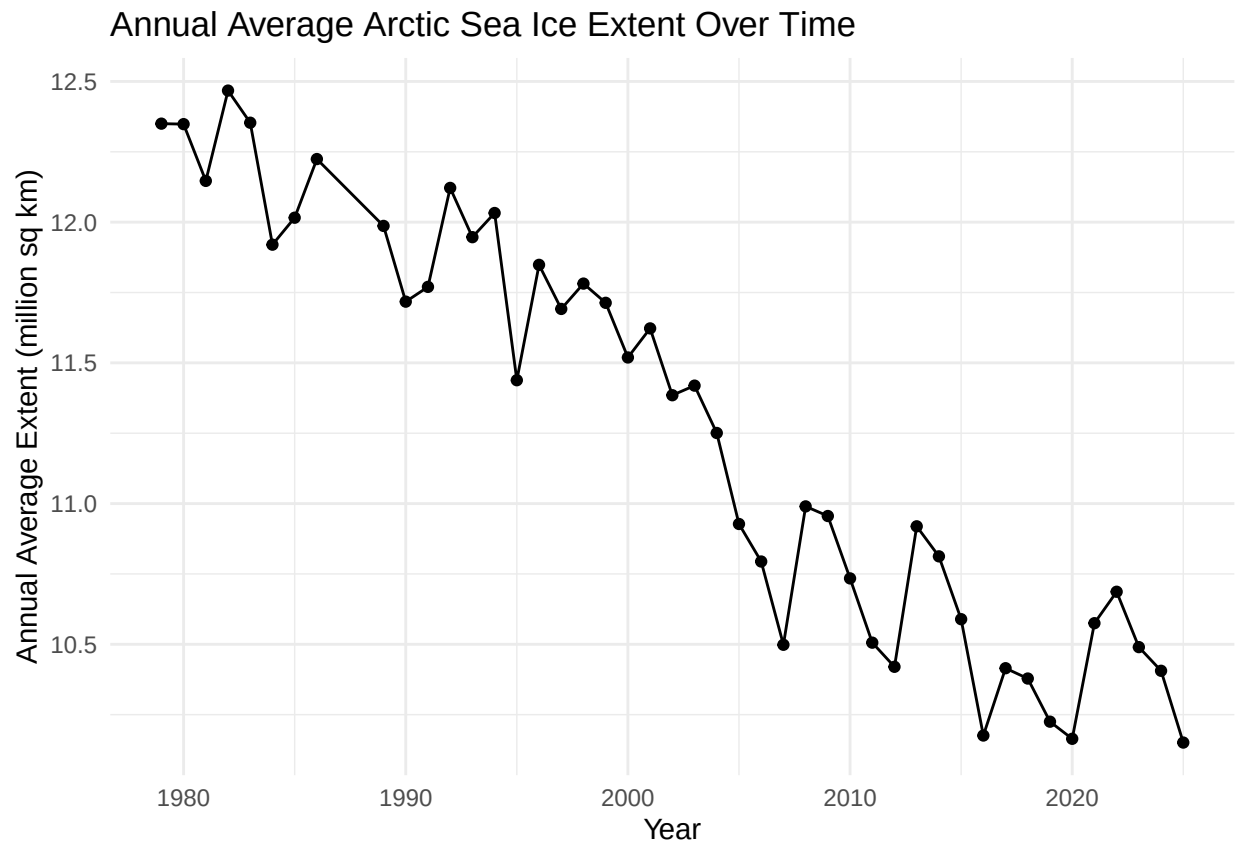


```

# Plot 2: Annual average sea ice extent over time
ggplot(annual_extent, aes(x = year, y = annual_extent)) +

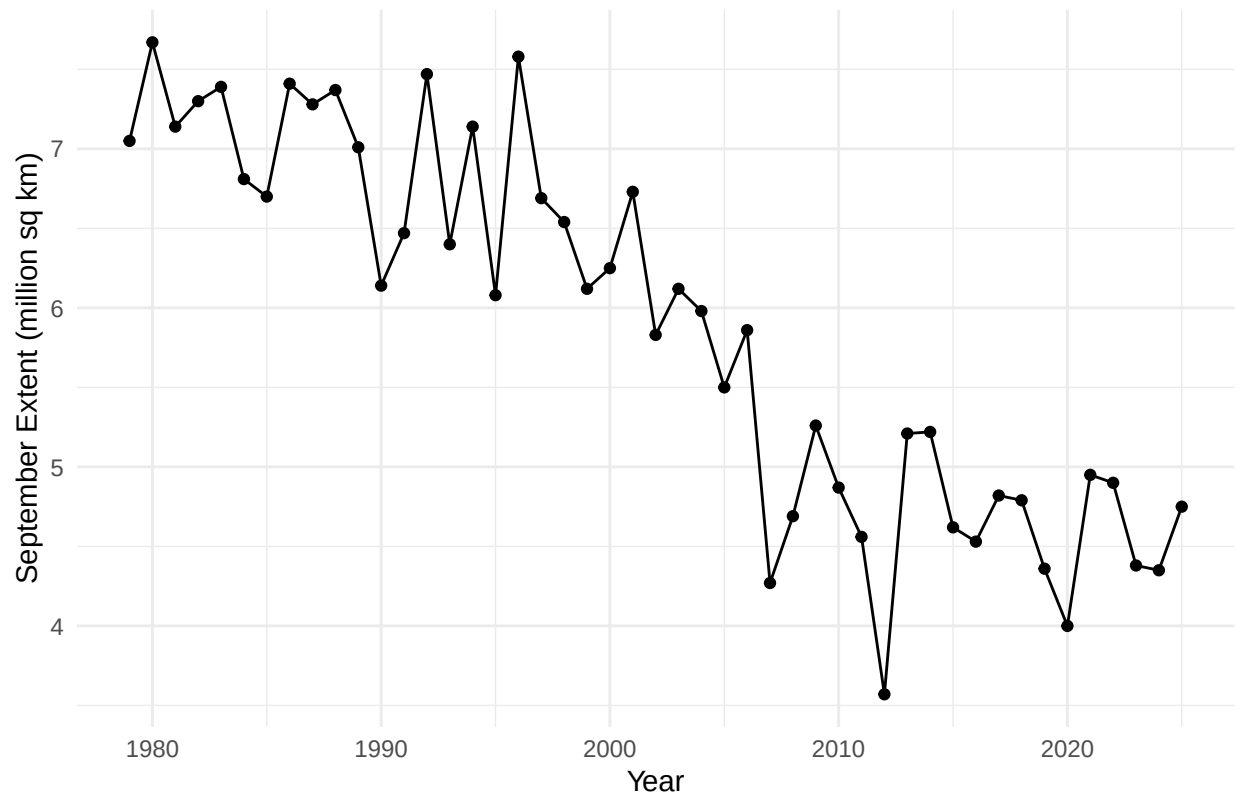
```

```
geom_line() +
geom_point() +
labs(
  title = "Annual Average Arctic Sea Ice Extent Over Time",
  x = "Year",
  y = "Annual Average Extent (million sq km)"
) +
theme_minimal()
```



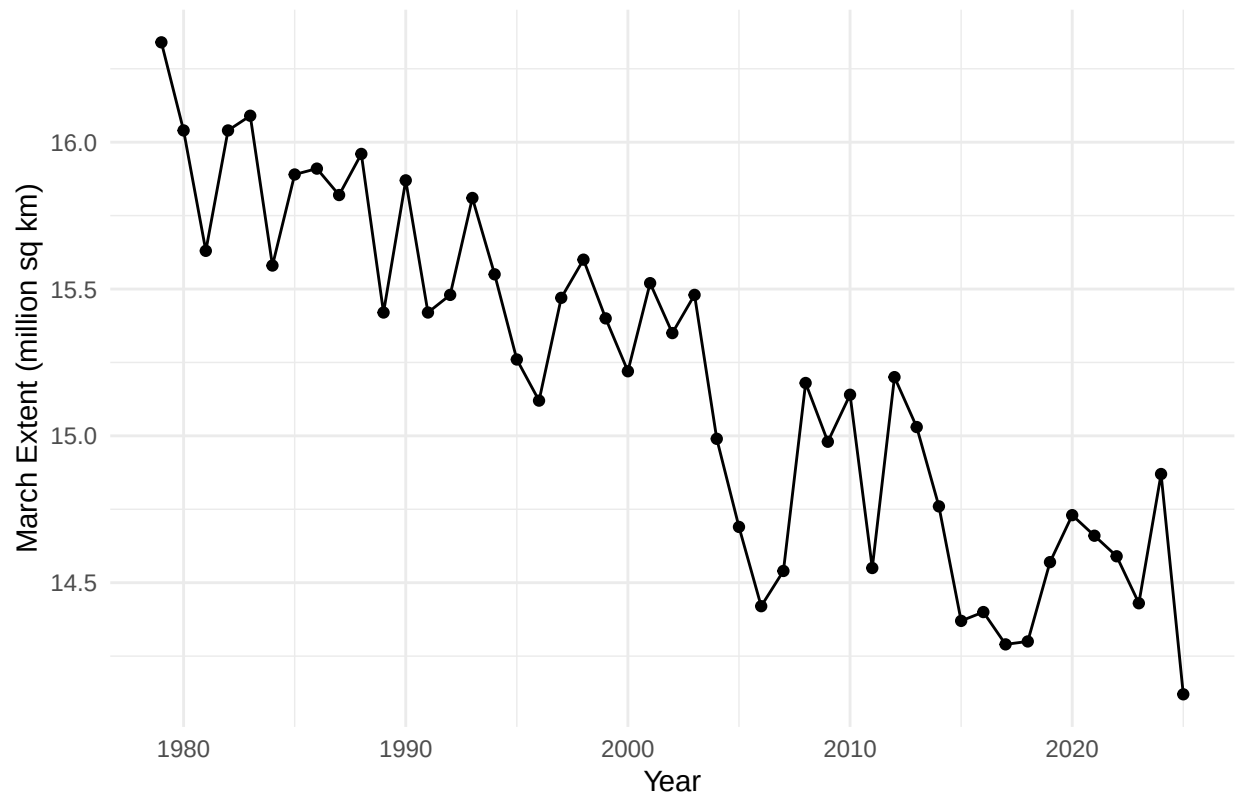
```
# Plot 3: September sea ice extent over time
ggplot(sept_data, aes(x = year, y = extent)) +
  geom_line() +
  geom_point() +
  labs(
    title = "September Arctic Sea Ice Extent Over Time",
    x = "Year",
    y = "September Extent (million sq km)"
  ) +
  theme_minimal()
```

September Arctic Sea Ice Extent Over Time

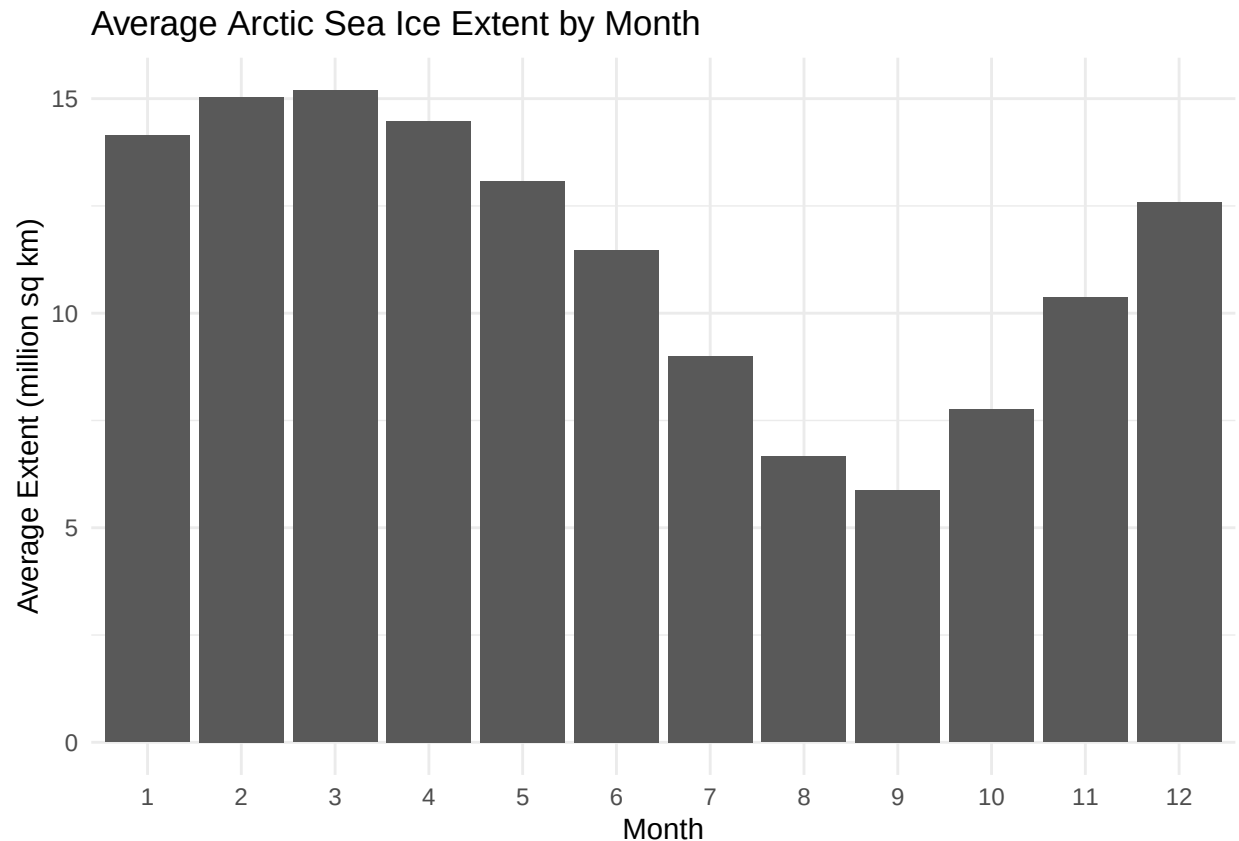


```
# Plot 4: March sea ice extent over time
ggplot(march_data, aes(x = year, y = extent)) +
  geom_line() +
  geom_point() +
  labs(
    title = "March Arctic Sea Ice Extent Over Time",
    x = "Year",
    y = "March Extent (million sq km)"
  ) +
  theme_minimal()
```


March Arctic Sea Ice Extent Over Time



```
# Plot 5: Average extent by month
ggplot(month_summary, aes(x = factor(month), y = mean_extent)) +
  geom_col() +
  labs(
    title = "Average Arctic Sea Ice Extent by Month",
    x = "Month",
    y = "Average Extent (million sq km)"
  ) +
  theme_minimal()
```



We can see there is a strong yearly seasonal cycle, a clear downward shift in the cycle over time, and large losses in late summer and early fall. Sea ice grows substantially in the winter, thus March has high levels, but in summer ice melts leading to September's ice falls faster. We will focus on September for forecasting near-zero conditions.

3: Linear Trend Analysis

```
# Fit a linear model using annual average extent over time
linear_trend <- lm(annual_extent ~ year, data = annual_extent)
```

```
# Print model summary
summary(linear_trend)
```

```
##
## Call:
## lm(formula = annual_extent ~ year, data = annual_extent)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.54429 -0.18952  0.02344  0.18162  0.40674
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 113.091706    5.107921    22.14    <2e-16 ***
## year        -0.050847    0.002551   -19.94    <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2311 on 43 degrees of freedom
## Multiple R-squared:  0.9024, Adjusted R-squared:  0.9001
## F-statistic: 397.4 on 1 and 43 DF,  p-value: < 2.2e-16
```

```
# Extract slope coefficient
linear_slope <- coef(linear_trend)["year"]
```

```
# Calculate decline per decade
decline_per_decade <- linear_slope * 10
```

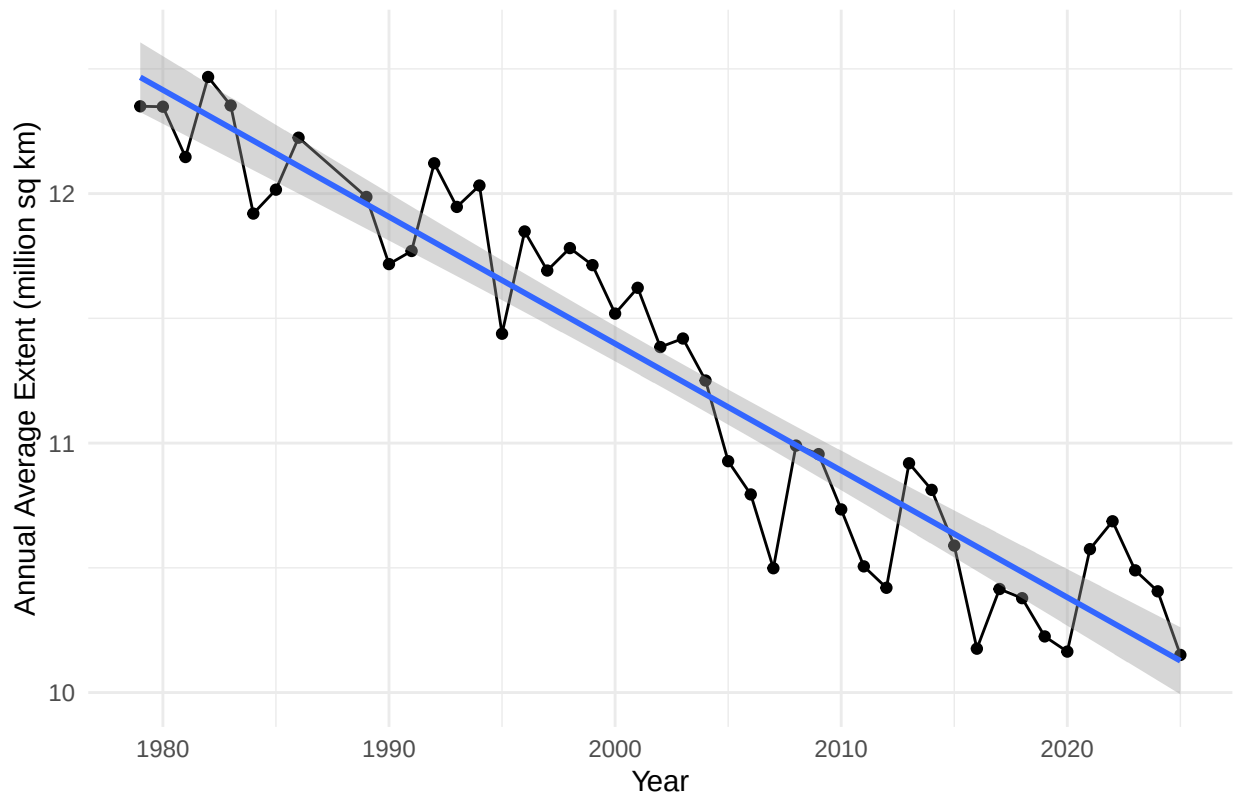
```
# Print decline per decade
decline_per_decade
```

```
##      year
## -0.5084658
```

```
# Plot annual extent with fitted linear trend line
ggplot(annual_extent, aes(x = year, y = annual_extent)) +
  geom_point() +
  geom_line() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Linear Trend in Annual Arctic Sea Ice Extent",
    x = "Year",
    y = "Annual Average Extent (million sq km)"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Linear Trend in Annual Arctic Sea Ice Extent



Annual extent = $113.091706 - 0.050847 \times \text{year}$

Slope estimate: -0.050847 million sq km per year

Translates to -0.5084658 million sq km per decade

Arctic sea ice extent has declined by approximately 0.508 million squares kilometers per decade on average. The decline is highly significant with a p-value < 0.05, indicating strong evidence that Arctic sea ice is decreasing over time.

63% of the variation in annual Arctic sea ice extent is explained by time alone (looking at R-squared).

The plot shows a clear downward trend over time with some year-to-year variability. Recent values are generally lower than earlier decades.

4: Seasonal Difference Analysis

```
# Fit a linear model for September extent over time
sept_linear <- lm(extent ~ year, data = sept_data)

# Print both summaries
summary(sept_linear)
```

```
##
## Call:
## lm(formula = extent ~ year, data = sept_data)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.54447 -0.30388  0.08585  0.30351  1.24851
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 158.154894   11.277010    14.03  <2e-16 ***
## year        -0.076064    0.005633   -13.50  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5238 on 45 degrees of freedom
## Multiple R-squared:  0.8021, Adjusted R-squared:  0.7977
## F-statistic: 182.4 on 1 and 45 DF,  p-value: < 2.2e-16

# Extract both slopes
sept_slope <- coef(sept_linear)["year"]
```

Here we see that September sea ice is declining by 0.76 million sq km per decade, or 0.076 per year. This is significant with a p-value < 0.05 and R-squared of 0.8.

```
# Fit a linear model for March extent over time
march_linear <- lm(extent ~ year, data = march_data)

# Print both summaries
summary(march_linear)
```

```
##
## Call:
## lm(formula = extent ~ year, data = march_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.6211 -0.2301  0.0527  0.2178  0.5131
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  91.296045   5.598630    16.31  <2e-16 ***
## year        -0.038013   0.002796   -13.59  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2601 on 45 degrees of freedom
## Multiple R-squared:  0.8042, Adjusted R-squared:  0.7998
## F-statistic: 184.8 on 1 and 45 DF,  p-value: < 2.2e-16

# Extract both slopes
march_slope <- coef(march_linear)["year"]
march_slope
```

```
##      year
## -0.03801341
```

```
# Compare slopes
```

```
slope_difference <- sept_slope - march_slope
```

```
slope_difference
```

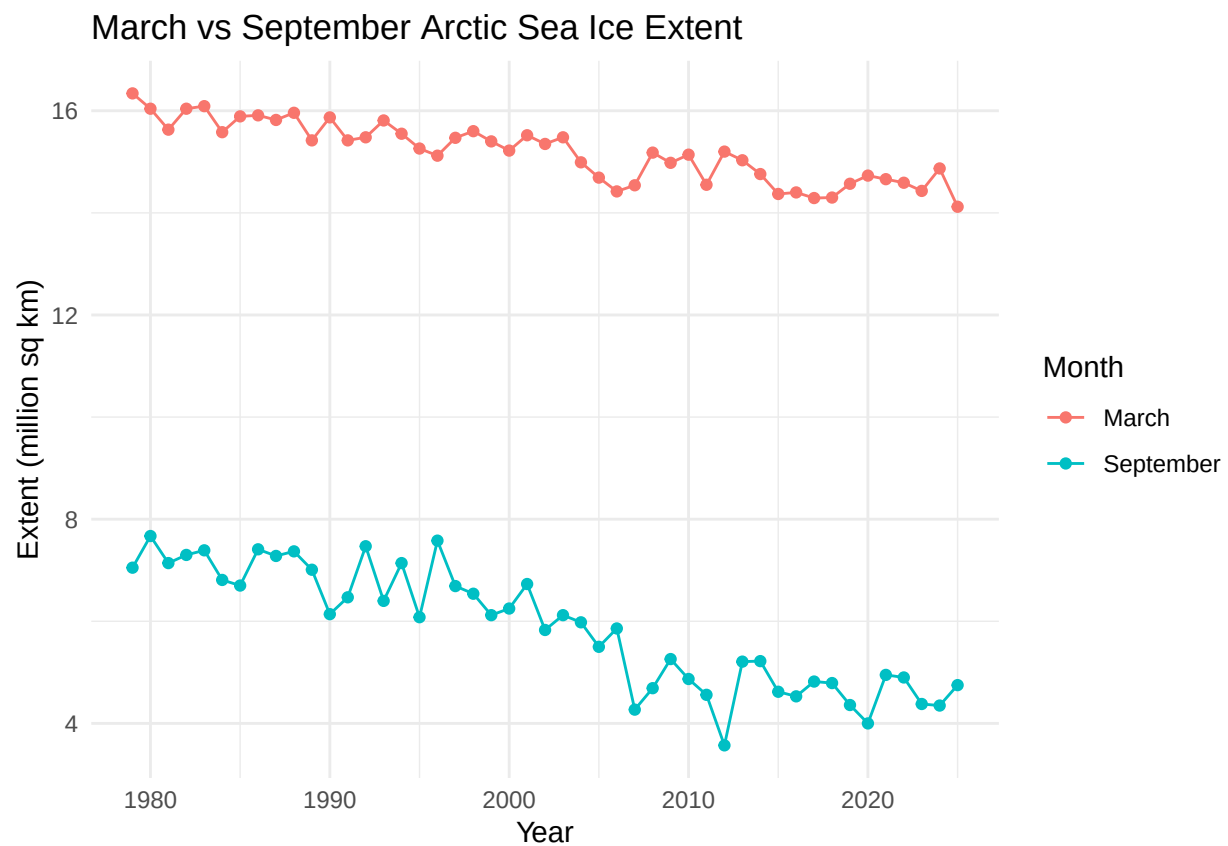
```
##      year
```

```
## -0.03805042
```

```
# Create a combined March vs September plot
```

```
march_sept_plot <- bind_rows(  
  march_data %>% mutate(season_group = "March"),  
  sept_data %>% mutate(season_group = "September")  
)
```

```
ggplot(march_sept_plot, aes(x = year, y = extent, color = season_group)) +  
  geom_line() +  
  geom_point() +  
  labs(  
    title = "March vs September Arctic Sea Ice Extent",  
    x = "Year",  
    y = "Extent (million sq km)",  
    color = "Month"  
  ) +  
  theme_minimal()
```



```
slope_difference
```

```
##          year  
## -0.03805042
```

The march slope is -0.0380 per year and -0.38 million sq km per decade with a significant p-value and R-squared of 0.8.

Overall, September decline is much steeper than March decline, it is declining about 2x as fast as March. The Arctic is losing ice much faster during the summer melt season. We see that there is a 0.03805042 difference between the September and march slopes, meaning September sea ice is declining 0.038 million sq km per year faster than March or 0.38 million sq km per decade. Thus, there is a large seasonal difference in this data as ice loss is concentrated in the summer.

5: Acceleration Test w/ Quadratic Model

```
# Fit a quadratic model using annual average extent  
quadratic_trend <- lm(annual_extent ~ year + I(year^2), data = annual_extent)
```

```
# Print model summary  
summary(quadratic_trend)
```

```
##  
## Call:  
## lm(formula = annual_extent ~ year + I(year^2), data = annual_extent)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.5437 -0.1894  0.0222  0.1819  0.4060   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  1.276e+02  8.318e+02   0.153   0.879      
## year         -6.532e-02  8.310e-01  -0.079   0.938      
## I(year^2)     3.615e-06  2.075e-04   0.017   0.986      
##  
## Residual standard error: 0.2338 on 42 degrees of freedom  
## Multiple R-squared:  0.9024, Adjusted R-squared:  0.8977   
## F-statistic: 194.1 on 2 and 42 DF,  p-value: < 2.2e-16
```

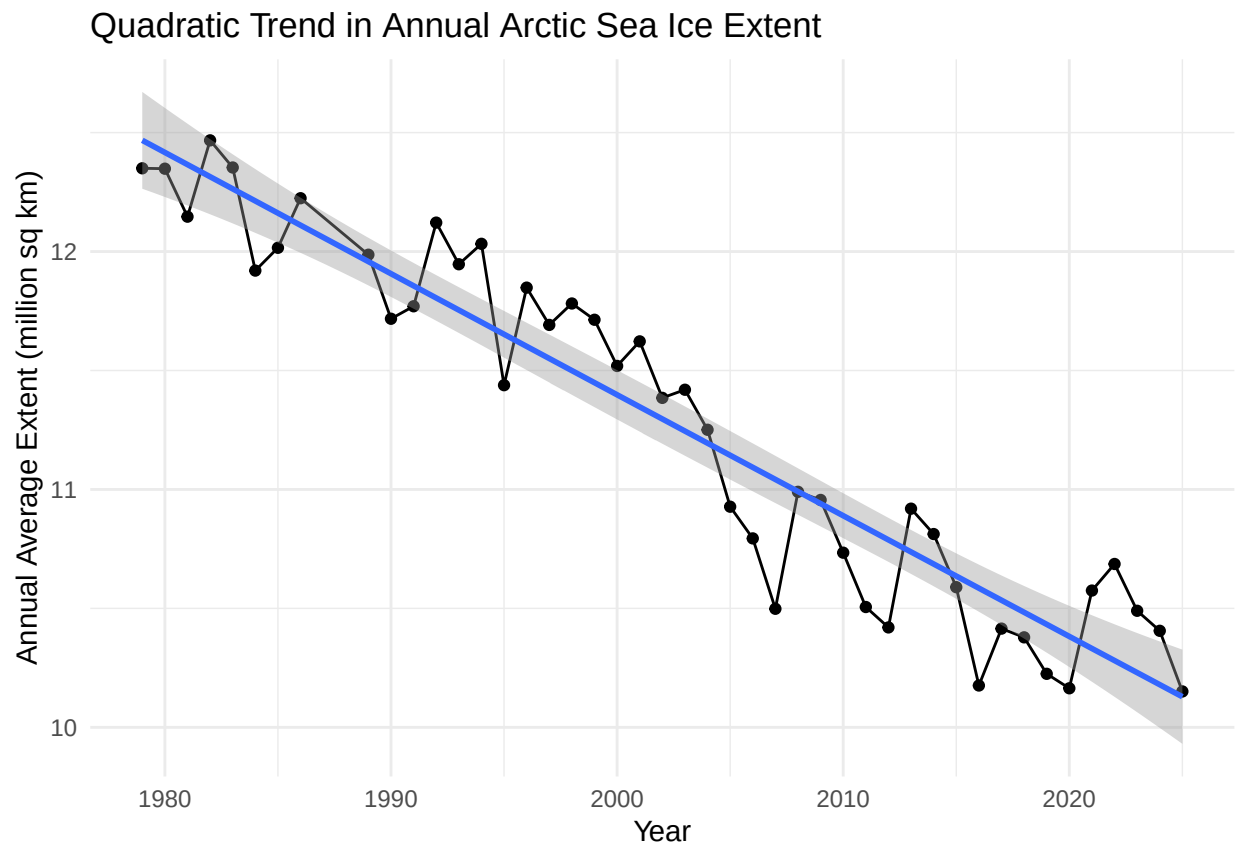
```
# Compare linear and quadratic models using AIC and BIC  
AIC(linear_trend, quadratic_trend)
```

```
##              df      AIC  
## linear_trend    3 -0.191596  
## quadratic_trend  4  1.808079
```

```
BIC(linear_trend, quadratic_trend)
```

```
##           df      BIC
## linear_trend    3 5.228391
## quadratic_trend  4 9.034729
```

```
# Plot annual extent with quadratic fit
ggplot(annual_extent, aes(x = year, y = annual_extent)) +
  geom_point() +
  geom_line() +
  stat_smooth(method = "lm", formula = y ~ x + I(x^2), se = TRUE) +
  labs(
    title = "Quadratic Trend in Annual Arctic Sea Ice Extent",
    x = "Year",
    y = "Annual Average Extent (million sq km)"
  ) +
  theme_minimal()
```



The quadratic model: $\text{extent} = \beta_0$

The β_2 term tells us if the trend is accelerating or decelerating. We see that the year coefficient has a value of -0.06532 and year² coefficient has a value of 0.000003615, both of which are statistically significant. The linear coefficient confirms that there is an overall decline of sea ice extent, the quadratic coefficient is positive and thus means that the decline is slowing down over time (not accelerating).

Looking at AIC and BIC, we see that the quadratic fits slightly better than the linear model. With the plot, we can see that the declining is slowly slightly with earlier years having a steeper decline and later years having a flatter trend.

6: Volatility Analysis

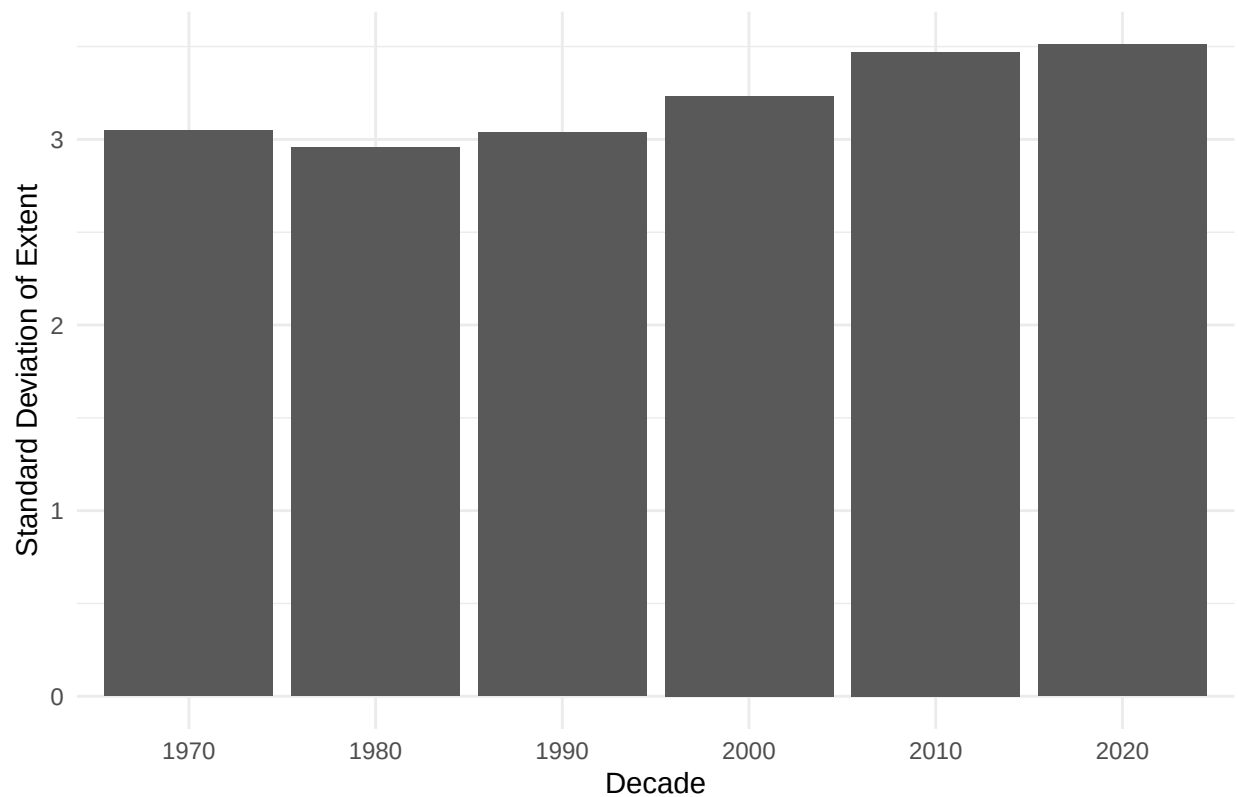
```
# Standard deviation of extent by decade
decade_volatility <- sea_ice %>%
  group_by(decade) %>%
  summarise(
    sd_extent = sd(extent, na.rm = TRUE)
  )

print(decade_volatility)
```

```
## # A tibble: 6 x 2
##   decade sd_extent
##   <dbl>     <dbl>
## 1  1970         3.05
## 2  1980         2.96
## 3  1990         3.04
## 4  2000         3.23
## 5  2010         3.47
## 6  2020         3.51
```

```
# Plot decade standard deviation
ggplot(decade_volatility, aes(x = factor(decade), y = sd_extent)) +
  geom_col() +
  labs(
    title = "Sea Ice Extent Volatility by Decade",
    x = "Decade",
    y = "Standard Deviation of Extent"
  ) +
  theme_minimal()
```

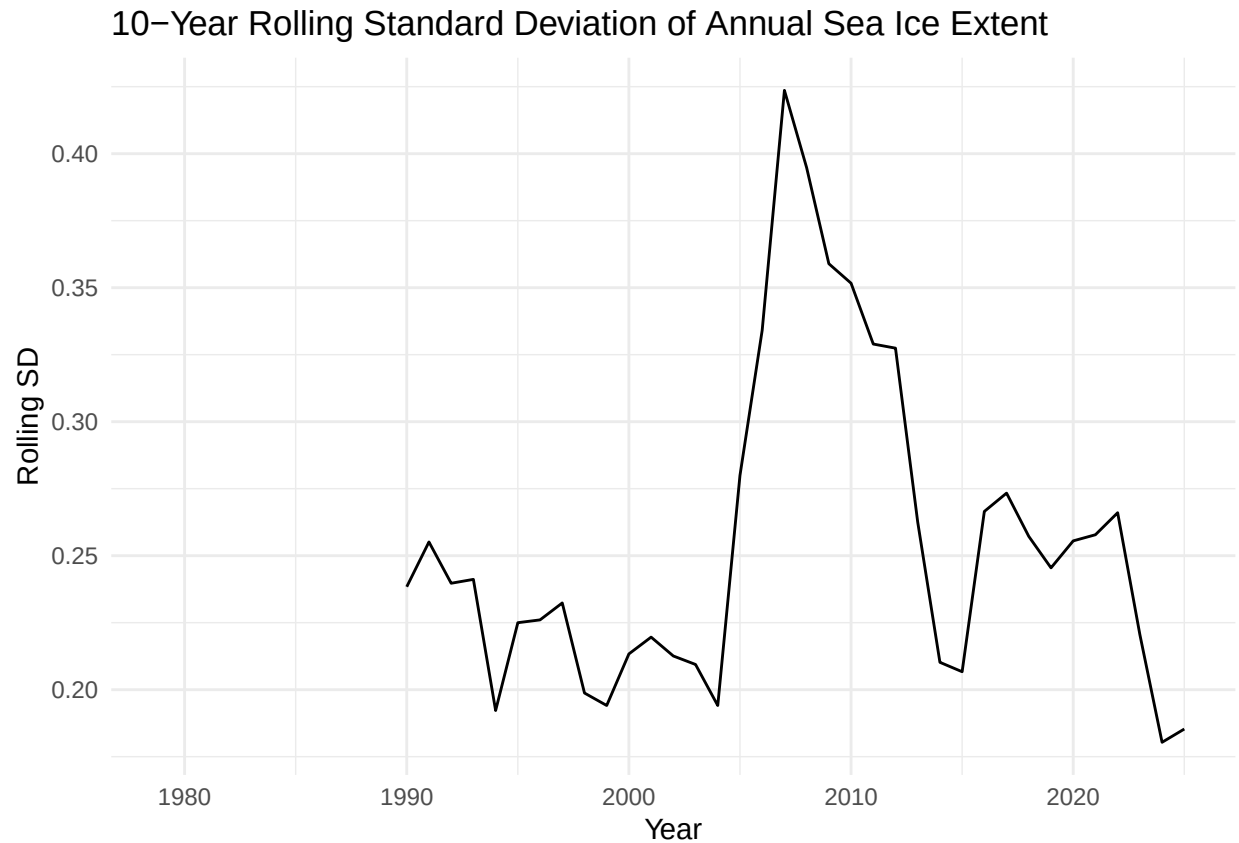
Sea Ice Extent Volatility by Decade



```
# Compute 10-year rolling standard deviation using annual averages
annual_extent <- annual_extent %>%
  arrange(year) %>%
  mutate(
    rolling_sd_10yr = rollapply(
      annual_extent,
      width = 10,
      FUN = sd,
      align = "right",
      fill = NA
    )
  )

# Plot rolling 10-year standard deviation
ggplot(annual_extent, aes(x = year, y = rolling_sd_10yr)) +
  geom_line() +
  labs(
    title = "10-Year Rolling Standard Deviation of Annual Sea Ice Extent",
    x = "Year",
    y = "Rolling SD"
  ) +
  theme_minimal()
```

```
## Warning: Removed 9 rows containing missing values or values outside the scale range
## ('geom_line()').
```



We see that volatility was fairly stable from the 1970s-1990s. Volatility began to increase in the 2000s and the highest variability occurs in the 2010s-2020s. Sea ice is declining and is becoming more unpredictable and variable year-to-year. Modern Arctic sea ice is more volatile than in earlier decades. There was a huge spike in the rolling SD around 2006-2012. This suggests more extreme events and greater uncertainty in future ice levels.

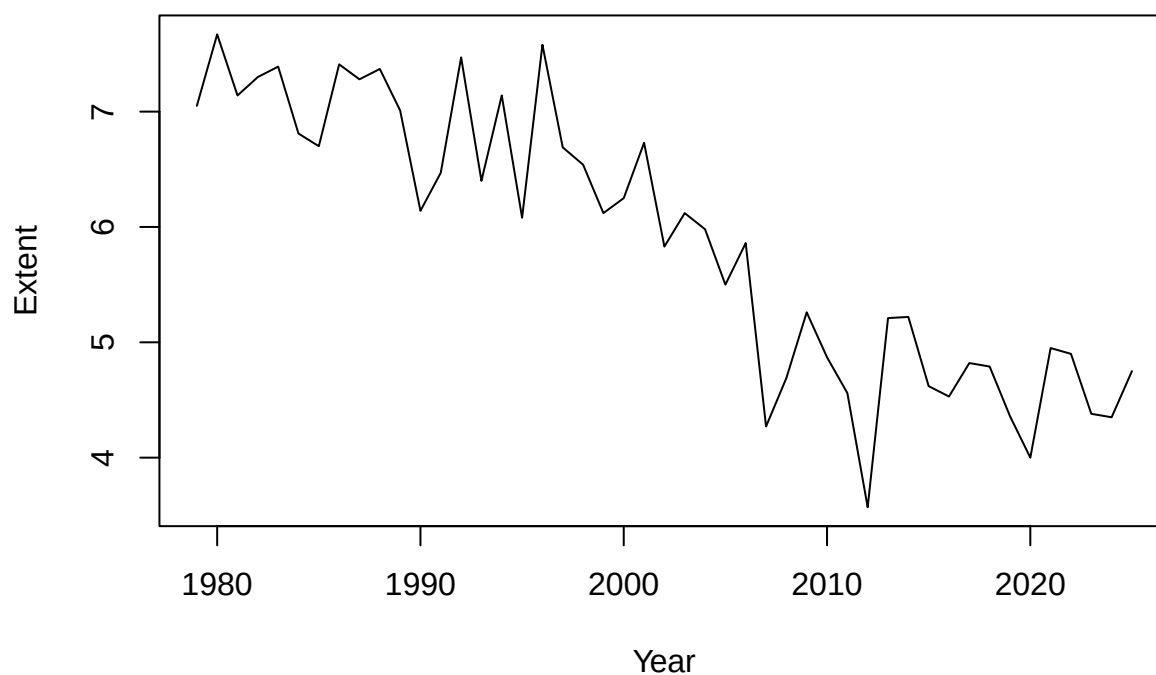
7: Prep Dataset for Forecasting

```
# Use September extent only for forecasting
forecast_data <- sept_data %>%
  select(year, extent) %>%
  arrange(year)

# Create time series object for September extent
# (Frequency = 1 because using yearly September data)
sept_ts <- ts(forecast_data$extent, start = min(forecast_data$year), frequency = 1)

# Plot September time series
plot(sept_ts, main = "September Arctic Sea Ice Extent Time Series", ylab = "Extent", xlab = "Year")
```

September Arctic Sea Ice Extent Time Series



```
# Run Augmented Dickey-Fuller test on the original September series
adf_original <- adf.test(sept_ts)
adf_original
```

```
##
## Augmented Dickey-Fuller Test
##
## data: sept_ts
## Dickey-Fuller = -1.7955, Lag order = 3, p-value = 0.6555
## alternative hypothesis: stationary
```

```
# Difference the series once if nonstationary
sept_ts_diff <- diff(sept_ts)
```

```
# Run the ADF test on the differenced series
adf_diff <- adf.test(sept_ts_diff)
```

```
## Warning in adf.test(sept_ts_diff): p-value smaller than printed p-value
```

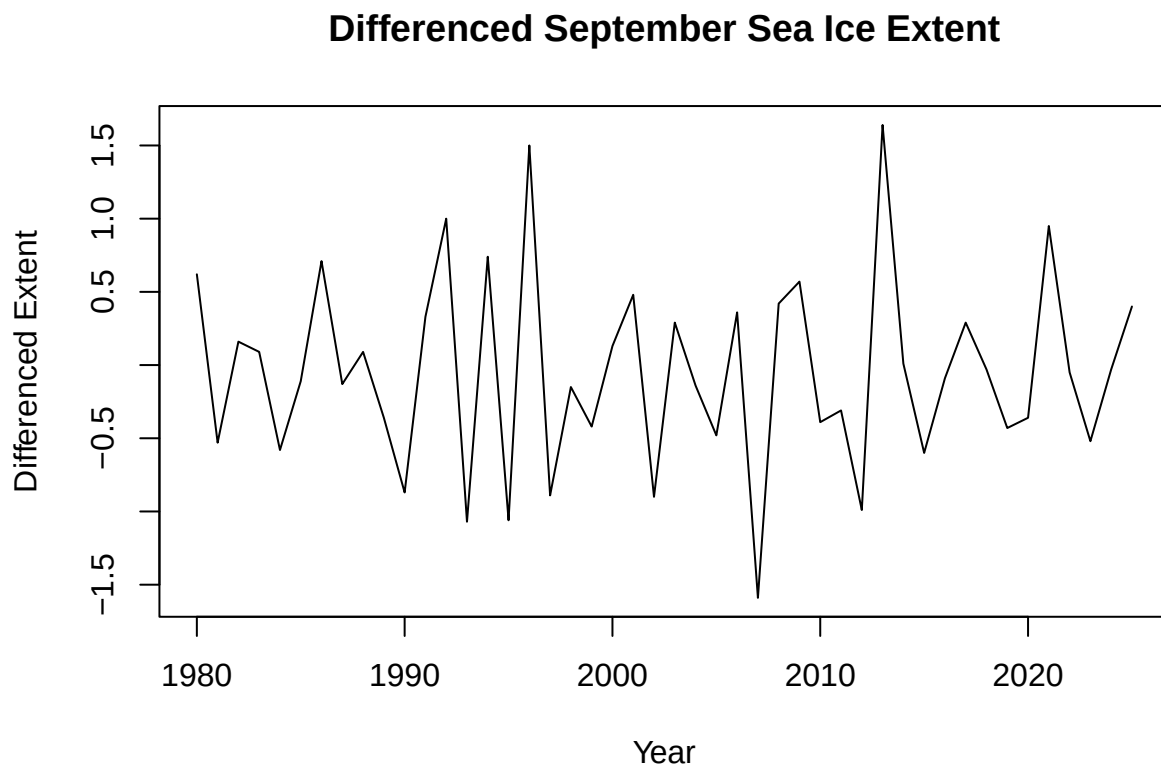
```
adf_diff
```

```
##
## Augmented Dickey-Fuller Test
##
```

```
## data: sept_ts_diff
## Dickey-Fuller = -6.2441, Lag order = 3, p-value = 0.01
## alternative hypothesis: stationary
```

```
# Plot the differenced series
```

```
plot(sept_ts_diff, main = "Differenced September Sea Ice Extent", ylab = "Differenced Extent", xlab = "Year")
```



From the non-differenced plot we see strong year-to-year variability and the ADF test concludes that the series is non-stationary, thus we cannot model this directly with ARIMA yet.

We apply a first difference to the September data to remove the trend and the ADF concludes that the data is now stationary. We now have data that captures year-to-year changes and is suitable for ARIMA modeling. We chose September data as it represents the annual minimum after summer melting, is most sensitive to climate change, and is best for forecasting near-zero.

8: Chronological Train/Test Split

```
# Create training data from 1979 to 2010
```

```
train_data <- forecast_data %>%
  filter(year >= 1979 & year <= 2010)
```

```
# Create testing data from 2011 onward
```

```
test_data <- forecast_data %>%
  filter(year >= 2011)
```

```
# Check sizes of train and test sets  
nrow(train_data)
```

```
## [1] 32
```

```
nrow(test_data)
```

```
## [1] 15
```

```
# Create time series objects for training and testing  
train_ts <- ts(train_data$extent, start = min(train_data$year), frequency = 1)  
test_ts <- ts(test_data$extent, start = min(test_data$year), frequency = 1)
```

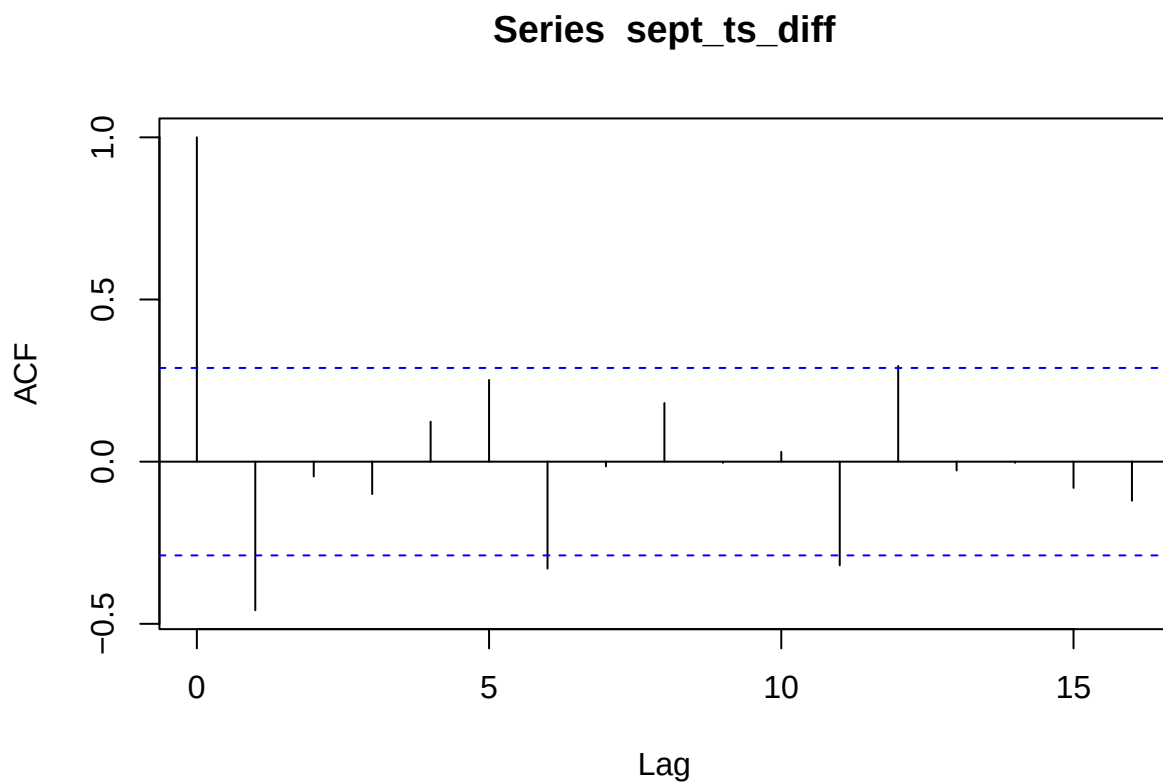
The dataset was split chronologically into a training set (1979-2010, where $n = 32$) and a testing set (2011-present, where $n = 15$). This ensures that model evaluation reflects the true out-of-sample forecasting and avoids data leakage.

9: ARIMA Model

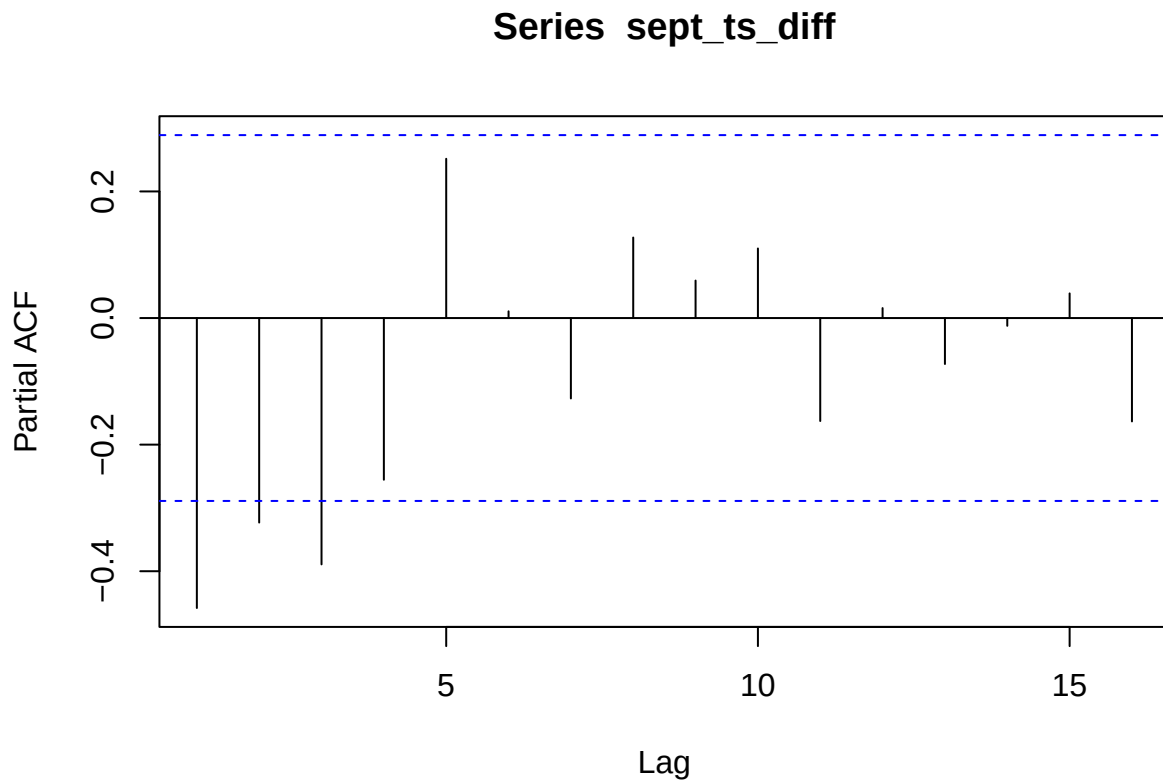
ACF/PACF plots, multiple candidate models, diagnostics filtering, final model selection

ACF & PACF

```
acf(sept_ts_diff)
```



```
pacf(sept_ts_diff)
```



ACF has spike at lag 1,6,11: MA(1) behavior PACF has spike at lag 1,3: MA process ARIMA(0,1,1) is the best model

Fit candidate models

```
model_011 <- arima(sept_ts, order = c(0,1,1))
model_111 <- arima(sept_ts, order = c(1,1,1))
model_010 <- arima(sept_ts, order = c(0,1,0))
model_110 <- arima(sept_ts, order = c(1,1,0))
```

AIC/BIC of models

```
AIC(model_011, model_111, model_010, model_110)
```

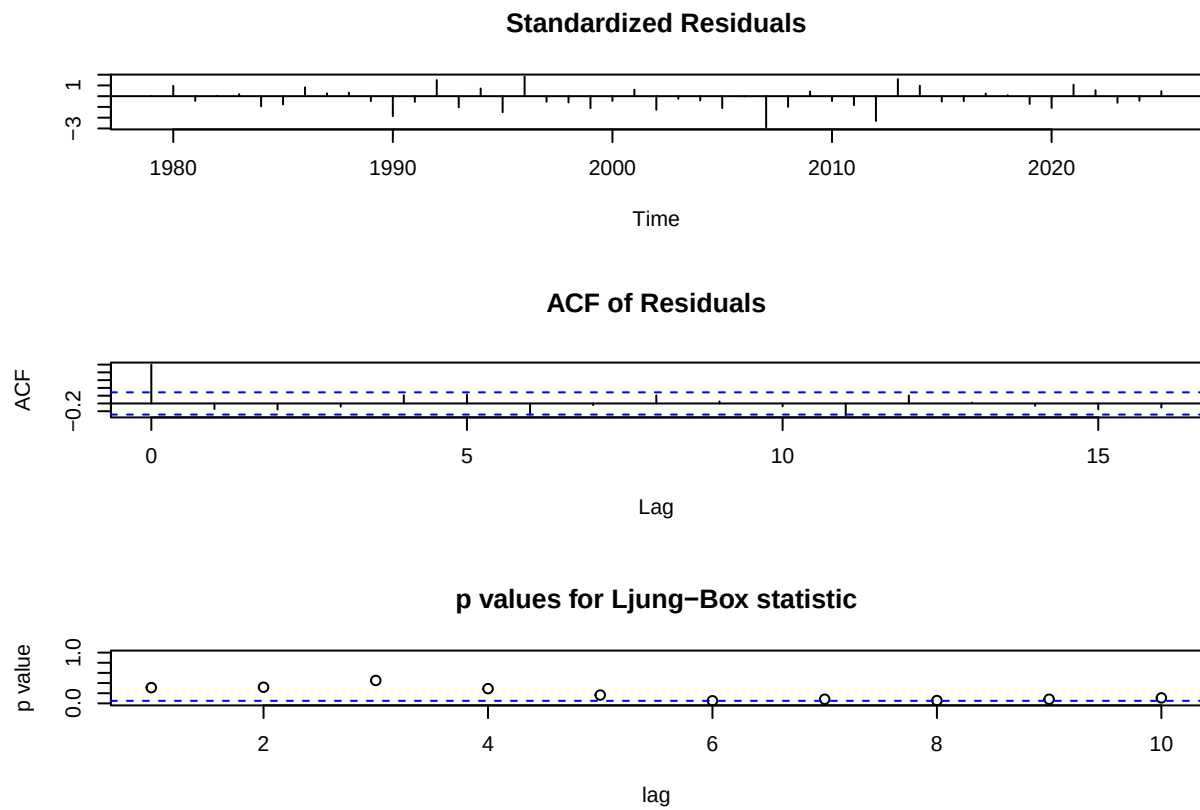
```
##          df      AIC
## model_011  2 80.07912
## model_111  3 81.79828
## model_010  1 94.68943
## model_110  2 86.33912
```

```
BIC(model_011, model_111, model_010, model_110)
```

```
##          df      BIC
## model_011  2 83.73640
## model_111  3 87.28420
## model_010  1 96.51807
## model_110  2 89.99640
```

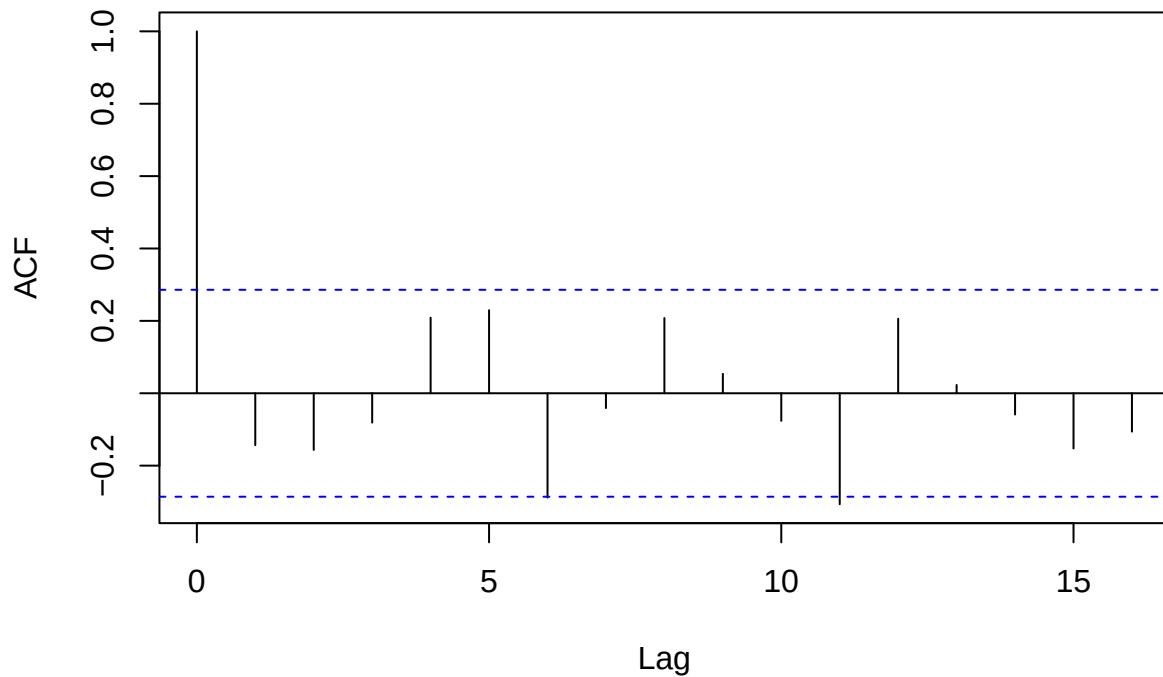
Diagnostics

```
tsdiag(model_011)
```



```
acf(residuals(model_011))
```


Series residuals(model_011)



```
Box.test(residuals(model_011), lag=10, type="Ljung")
```

```
##  
## Box-Ljung test  
##  
## data: residuals(model_011)  
## X-squared = 15.716, df = 10, p-value = 0.1081
```

The residuals fluctuate around 0 with no clear trend, most residual autocorrelations are within the bonds, and the Ljung-Box test gives a p-value of 0.1081 which is greater than 0.05. Thus, we fail to reject the null of no autocorrelation, there is no strong evidence that the residuals contain time dependence; this is ideally what we want for a good ARIMA model.

10: Fit Forecasting Models

```
### MODEL 1: LINEAR TREND FORECAST  
  
# Fit linear model on September training data  
linear_forecast_model <- lm(extent ~ year, data = train_data)  
  
# Predict on the test set with 80% prediction intervals  
linear_pred <- predict(
```

```

linear_forecast_model,
newdata = test_data,
interval = "prediction",
level = 0.80
)

# Convert predictions to a data frame
linear_results <- data.frame(
  year = test_data$year,
  actual = test_data$extent,
  fit = linear_pred[, "fit"],
  lwr = linear_pred[, "lwr"],
  upr = linear_pred[, "upr"]
)

### MODEL 2: QUADRATIC TREND FORECAST

# Fit quadratic model on September training data
quadratic_forecast_model <- lm(extent ~ year + I(year^2), data = train_data)

# Predict on the test set with 80% prediction intervals
quadratic_pred <- predict(
  quadratic_forecast_model,
  newdata = test_data,
  interval = "prediction",
  level = 0.80
)

# Convert predictions to a data frame
quadratic_results <- data.frame(
  year = test_data$year,
  actual = test_data$extent,
  fit = quadratic_pred[, "fit"],
  lwr = quadratic_pred[, "lwr"],
  upr = quadratic_pred[, "upr"]
)

### MODEL 3: ARIMA FORECAST

# Fit ARIMA model automatically on September training data
arima_model_011 <- arima(train_ts, order = c(0,1,1))

# Print ARIMA model details
summary(arima_model_011)

##
## Call:
## arima(x = train_ts, order = c(0, 1, 1))
##
## Coefficients:
##          ma1
##        -0.5744
## s.e.    0.1175

```

```
##
## sigma^2 estimated as 0.3133: log likelihood = -26.2, aic = 56.4
##
## Training set error measures:
##           ME      RMSE      MAE      MPE      MAPE      MASE
## Training set -0.1583249 0.5509521 0.4335565 -3.30728 7.131806 0.7832314
##           ACF1
## Training set -0.2222644

# Forecast for the length of the test period with 80% intervals
arima_fc_update <- forecast(arima_model_011, h = length(test_ts), level = 80)

# Convert ARIMA forecast output to a data frame
arima_results <- data.frame(
  year   = test_data$year,
  actual = as.numeric(test_ts),
  fit    = as.numeric(arima_fc_update$mean),
  lwr    = as.numeric(arima_fc_update$lower[, 1]),
  upr    = as.numeric(arima_fc_update$upper[, 1])
)

linear_results
```

```
##   year actual      fit      lwr      upr
## 1  2011   4.56 5.210746 4.492236 5.929256
## 2  2012   3.57 5.134276 4.411819 5.856733
## 3  2013   5.21 5.057806 4.331192 5.784420
## 4  2014   5.22 4.981336 4.250360 5.712312
## 5  2015   4.62 4.904866 4.169326 5.640406
## 6  2016   4.53 4.828396 4.088095 5.568698
## 7  2017   4.82 4.751926 4.006669 5.497183
## 8  2018   4.79 4.675456 3.925053 5.425859
## 9  2019   4.36 4.598986 3.843251 5.354722
## 10 2020   4.00 4.522516 3.761266 5.283767
## 11 2021   4.95 4.446047 3.679103 5.212990
## 12 2022   4.90 4.369577 3.596765 5.142388
## 13 2023   4.38 4.293107 3.514257 5.071956
## 14 2024   4.35 4.216637 3.431582 5.001692
## 15 2025   4.75 4.140167 3.348744 4.931590
```

```
quadratic_results
```

```
##   year actual      fit      lwr      upr
## 1  2011   4.56 4.591532 3.90972440 5.273340
## 2  2012   3.57 4.402478 3.69849176 5.106464
## 3  2013   5.21 4.206801 3.47678409 4.936818
## 4  2014   5.22 4.004502 3.24445748 4.764546
## 5  2015   4.62 3.795580 3.00142003 4.589739
## 6  2016   4.53 3.580035 2.74762685 4.412443
## 7  2017   4.82 3.357868 2.48307300 4.232662
## 8  2018   4.79 3.129078 2.20778544 4.050370
## 9  2019   4.36 2.893665 1.92181494 3.865516
## 10 2020   4.00 2.651630 1.62522862 3.678032
```

```
## 11 2021 4.95 2.402972 1.31810371 3.487841
## 12 2022 4.90 2.147692 1.00052246 3.294862
## 13 2023 4.38 1.885789 0.67256831 3.099010
## 14 2024 4.35 1.617264 0.33432313 2.900204
## 15 2025 4.75 1.342116 -0.01413462 2.698366
```

```
arima_results
```

```
##   year actual      fit      lwr      upr
## 1  2011  4.56 4.998373 4.281004 5.715743
## 2  2012  3.57 4.998373 4.218742 5.778004
## 3  2013  5.21 4.998373 4.161098 5.835648
## 4  2014  5.22 4.998373 4.107175 5.889572
## 5  2015  4.62 4.998373 4.056333 5.940414
## 6  2016  4.53 4.998373 4.008098 5.988649
## 7  2017  4.82 4.998373 3.962106 6.034641
## 8  2018  4.79 4.998373 3.918070 6.078677
## 9  2019  4.36 4.998373 3.875760 6.120987
## 10 2020  4.00 4.998373 3.834988 6.161759
## 11 2021  4.95 4.998373 3.795597 6.201150
## 12 2022  4.90 4.998373 3.757456 6.239291
## 13 2023  4.38 4.998373 3.720452 6.276294
## 14 2024  4.35 4.998373 3.684491 6.312256
## 15 2025  4.75 4.998373 3.649488 6.347259
```

The linear trend forecast assumes sea ice declines at a constant rate each year. It consistently predicts values higher than observed, thus underestimates the rate of decline and overestimates ice.

The quadratic trend forecast allows for acceleration in decline. The quadratic model declines too aggressively and overestimates acceleration in ice loss.

The ARIMA model produces stable forecasts but doesn't force strong downward extrapolation. The model assumes future values follow recent dynamics. Overall, it is more conservative and stable.

11: Model Evaluation/Selection

```
# Create a helper function to calculate RMSE
rmse_fun <- function(actual, predicted) {
  sqrt(mean((actual - predicted)^2, na.rm = TRUE))
}

# Create a helper function to calculate MAE
mae_fun <- function(actual, predicted) {
  mean(abs(actual - predicted), na.rm = TRUE)
}

# Create a helper function to calculate interval coverage
coverage_fun <- function(actual, lower, upper) {
  mean(actual >= lower & actual <= upper, na.rm = TRUE)
}
```

```

# Evaluate linear model
linear_rmse <- rmse_fun(linear_results$actual, linear_results$fit)
linear_mae <- mae_fun(linear_results$actual, linear_results$fit)
linear_cov <- coverage_fun(linear_results$actual, linear_results$lwr, linear_results$upr)

# Evaluate quadratic model
quadratic_rmse <- rmse_fun(quadratic_results$actual, quadratic_results$fit)
quadratic_mae <- mae_fun(quadratic_results$actual, quadratic_results$fit)
quadratic_cov <- coverage_fun(quadratic_results$actual, quadratic_results$lwr, quadratic_results$upr)

# Evaluate ARIMA model
arima_rmse <- rmse_fun(arima_results$actual, arima_results$fit)
arima_mae <- mae_fun(arima_results$actual, arima_results$fit)
arima_cov <- coverage_fun(arima_results$actual, arima_results$lwr, arima_results$upr)

# Create a model comparison table
model_comparison <- data.frame(
  Model = c("Linear", "Quadratic", "ARIMA"),
  AIC = c(AIC(linear_forecast_model), AIC(quadratic_forecast_model), AIC(arima_model_011)),
  BIC = c(BIC(linear_forecast_model), BIC(quadratic_forecast_model), BIC(arima_model_011)),
  RMSE = c(linear_rmse, quadratic_rmse, arima_rmse),
  MAE = c(linear_mae, quadratic_mae, arima_mae),
  Coverage_80 = c(linear_cov, quadratic_cov, arima_cov)
)

print(model_comparison)

```

```

##      Model      AIC      BIC      RMSE      MAE Coverage_80
## 1   Linear 52.34730 56.74451 0.5416755 0.3998486 0.93333333
## 2 Quadratic 44.92431 50.78725 1.8795363 1.6486013 0.06666667
## 3    ARIMA 56.40000 59.26798 0.5799340 0.4554737 0.93333333

```

```

best_rmse_model <- model_comparison$Model[which.min(model_comparison$RMSE)]
best_rmse_model

```

```
## [1] "Linear"
```

The best model is the linear model as it has the lowest RMSE (prediction error), MAE (average error), has the best coverage ratio (0.933 with wide intervals as 0.8 is the targetted; thus they are safe but conservative). The quadratic model has the lowest AIC/BIC but performs the worst on the test data, it fails to generalize. The ARIMA performs similarly and better captures short-term variability, but its predictive accuracy is slightly lower. The linear model offers the best balance of predictive performance and simplicity for forecasting.

12: Refit and Forecast with Best Model

```

# Define how many years into the future to forecast
future_horizon <- 50

```

```

# Define near-zero threshold
near_zero_threshold <- 1

### Final linear model forecast
final_linear_model <- lm(extent ~ year, data = forecast_data)

# Create future years for forecasting
future_years <- data.frame(
  year = seq(max(forecast_data$year) + 1, max(forecast_data$year) + future_horizon, by = 1)
)

# Forecast future values from final linear model
final_linear_pred <- predict(
  final_linear_model,
  newdata = future_years,
  interval = "prediction",
  level = 0.80
)

# Store linear forecast results
final_linear_results <- data.frame(
  year = future_years$year,
  fit = final_linear_pred[, "fit"],
  lwr = final_linear_pred[, "lwr"],
  upr = final_linear_pred[, "upr"]
)
final_linear_results

```

##	year	fit	lwr	upr
## 1	2026	4.0495745	3.33896859	4.760180
## 2	2027	3.9735106	3.26105663	4.685965
## 3	2028	3.8974468	3.18307431	4.611819
## 4	2029	3.8213830	3.10502220	4.537744
## 5	2030	3.7453191	3.02690088	4.463737
## 6	2031	3.6692553	2.94871094	4.389800
## 7	2032	3.5931915	2.87045299	4.315930
## 8	2033	3.5171277	2.79212764	4.242128
## 9	2034	3.4410638	2.71373552	4.168392
## 10	2035	3.3650000	2.63527728	4.094723
## 11	2036	3.2889362	2.55675356	4.021119
## 12	2037	3.2128723	2.47816502	3.947580
## 13	2038	3.1368085	2.39951232	3.874105
## 14	2039	3.0607447	2.32079614	3.800693
## 15	2040	2.9846809	2.24201716	3.727345
## 16	2041	2.9086170	2.16317607	3.654058
## 17	2042	2.8325532	2.08427355	3.580833
## 18	2043	2.7564894	2.00531030	3.507668
## 19	2044	2.6804255	1.92628703	3.434564
## 20	2045	2.6043617	1.84720443	3.361519
## 21	2046	2.5282979	1.76806321	3.288533
## 22	2047	2.4522340	1.68886409	3.215604
## 23	2048	2.3761702	1.60960777	3.142733
## 24	2049	2.3001064	1.53029497	3.069918

```
## 25 2050 2.2240426 1.45092639 2.997159
## 26 2051 2.1479787 1.37150275 2.924455
## 27 2052 2.0719149 1.29202476 2.851805
## 28 2053 1.9958511 1.21249313 2.779209
## 29 2054 1.9197872 1.13290858 2.706666
## 30 2055 1.8437234 1.05327180 2.634175
## 31 2056 1.7676596 0.97358350 2.561736
## 32 2057 1.6915957 0.89384439 2.489347
## 33 2058 1.6155319 0.81405516 2.417009
## 34 2059 1.5394681 0.73421651 2.344720
## 35 2060 1.4634043 0.65432913 2.272479
## 36 2061 1.3873404 0.57439371 2.200287
## 37 2062 1.3112766 0.49441093 2.128142
## 38 2063 1.2352128 0.41438147 2.056044
## 39 2064 1.1591489 0.33430600 1.983992
## 40 2065 1.0830851 0.25418520 1.911985
## 41 2066 1.0070213 0.17401972 1.840023
## 42 2067 0.9309574 0.09381022 1.768105
## 43 2068 0.8548936 0.01355735 1.696230
## 44 2069 0.7788298 -0.06673825 1.624398
## 45 2070 0.7027660 -0.14707593 1.552608
## 46 2071 0.6267021 -0.22745506 1.480859
## 47 2072 0.5506383 -0.30787503 1.409152
## 48 2073 0.4745745 -0.38833520 1.337484
## 49 2074 0.3985106 -0.46883497 1.265856
## 50 2075 0.3224468 -0.54937374 1.194267
```

```
library(forecast)

# Fit final ARIMA model on full September dataset
final_arima_model <- arima(sept_ts, order = c(0, 1, 1))

# Forecast future values using the ARIMA(0,1,1) model
final_arima_fc <- forecast(
  final_arima_model,
  h = future_horizon,
  level = 80
)

# Create data frame with forecast results
final_arima_results <- data.frame(
  year = seq(
    max(forecast_data$year) + 1,
    max(forecast_data$year) + future_horizon,
    by = 1
  ),
  fit = as.numeric(final_arima_fc$mean), # point forecast
  lwr = as.numeric(final_arima_fc$lower[, 1]), # lower bound (80%)
  upr = as.numeric(final_arima_fc$upper[, 1]) # upper bound (80%)
)

head(final_arima_results)
```

```
##   year    fit    lwr    upr
## 1 2026 4.593254 3.887783 5.298724
```

```
## 2 2027 4.593254 3.834943 5.351564
## 3 2028 4.593254 3.785552 5.400955
## 4 2029 4.593254 3.739013 5.447495
## 5 2030 4.593254 3.694881 5.491626
## 6 2031 4.593254 3.652818 5.533690
```

A linear model assumes that decline is inevitable while for an ARIMA decline is uncertain. The linear model forecasts a continued steady decline in Arctic sea ice extent while the ARIMA produces a flat forecast. The linear model provides a more realistic long-term projection while the ARIMA is better for short-term forecasting.

“A consensus has emerged among scientists that the Arctic Ocean is effectively ice free when its sea ice extent falls below 1 million square kilometers (390,000 square miles).” -NSIDC: <https://nsidc.org/learn/ask-scientist/what-do-scientists-mean-ice-free-arctic>

13: Calculate Near-Zero

```
# Find the first year below threshold for linear forecast
near_zero_year_linear <- final_linear_results %>%
  filter(fit < near_zero_threshold) %>%
  slice(1)

print(near_zero_year_linear)
```

```
##   year      fit      lwr      upr
## 42 2067 0.9309574 0.09381022 1.768105
```

```
# FOR ARIMA
near_zero_year_arima <- final_arima_results %>%
  filter(fit < near_zero_threshold) %>%
  slice(1)

print(near_zero_year_arima)
```

```
## [1] year fit lwr upr
## <0 rows> (or 0-length row.names)
```

According to the linear model, Arctic sea ice extent falls below the near-zero threshold (1 million sq km) around 2067 with a 0.09 lower bound (no ice) and 1.77 upper bound (slightly above threshold). The ARIMA never predicts the sea ice dropping below 1.

```
# Smallest forecasted value for linear
min(final_linear_results$fit)
```

```
## [1] 0.3224468
```

```
# Last forecasted years
tail(final_linear_results)
```



```
##   year      fit      lwr      upr
## 45 2070 0.7027660 -0.1470759 1.552608
## 46 2071 0.6267021 -0.2274551 1.480859
## 47 2072 0.5506383 -0.3078750 1.409152
## 48 2073 0.4745745 -0.3883352 1.337484
## 49 2074 0.3985106 -0.4688350 1.265856
## 50 2075 0.3224468 -0.5493737 1.194267
```

```
# Smallest predicted value for ARIMA
min(final_arima_results$fit)
```

```
## [1] 4.593254
```

```
# Last forecast values
tail(final_arima_results)
```

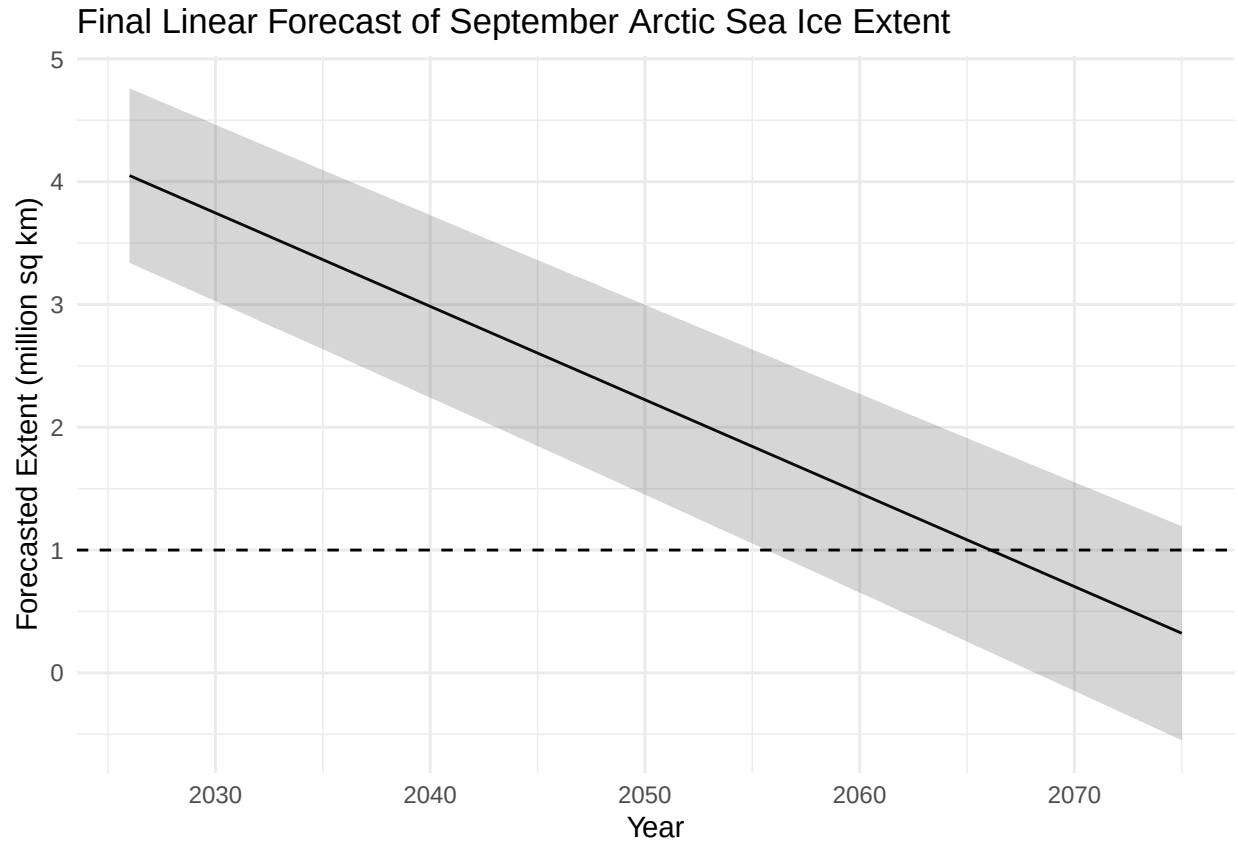
```
##   year      fit      lwr      upr
## 45 2070 4.593254 2.618179 6.568329
## 46 2071 4.593254 2.598694 6.587813
## 47 2072 4.593254 2.579398 6.607109
## 48 2073 4.593254 2.560285 6.626222
## 49 2074 4.593254 2.541351 6.645157
## 50 2075 4.593254 2.522589 6.663918
```

In the linear model we find the smallest predicted value for ice is at the end of the forecast period in 2075, by the 2070's ice is consistently near-zero. The smallest value for the ARIMA is at the end of the forecast period around 2075 at a value of 4.593, thus it predicts no long-term decline. The linear estimate provides the best and most accurate forecast of future Arctic sea ice conditions.

ARIMA(0,1,1) is a difference-based forecasting model. In many cases, once it learns the recent pattern in changes, its long-run forecast tends to level off rather than keep dropping linearly forever.

14: Plot Forecast

```
# Plot linear future forecast
ggplot(final_linear_results, aes(x = year, y = fit)) +
  geom_line() +
  geom_ribbon(aes(ymin = lwr, ymax = upr), alpha = 0.2) +
  geom_hline(yintercept = near_zero_threshold, linetype = "dashed") +
  labs(
    title = "Final Linear Forecast of September Arctic Sea Ice Extent",
    x = "Year",
    y = "Forecasted Extent (million sq km)"
  ) +
  theme_minimal()
```



We see that forecasted extent falls below 1 by 2065-2070, precisely at around 2067. The shaded region represents a 80% prediction interval, there are tighter bands in earlier years and wider bands in later years as long-term forecasts are less certain. This shows that near-zero conditions are very likely, even when accounting for uncertainty.